NAMMA SPACE COMPETITION • ROUND 1 SUBMISSION DOSSIER

RAASTA-3D

End-to-End Photorealistic 3D Digital Twin Reconstruction & Obstacle-Avoiding Indoor Navigation Engine

Organized by: Techfest, IIT Bombay in Partnership with Juspay

Primary Evaluation Space: Smart Home / Accessible Indoor Digital Twin

Submission Portal: Round 1 Complete Technical Package

Live Web Viewer: `https://raasta-3d.vercel.app`

Open Source Repository: `https://github.com/Sarthakadhav/Raasta-3D`

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SECTION 1: TECHNICAL ABSTRACT & SYSTEM ARCHITECTURE

TECHNICAL ABSTRACT

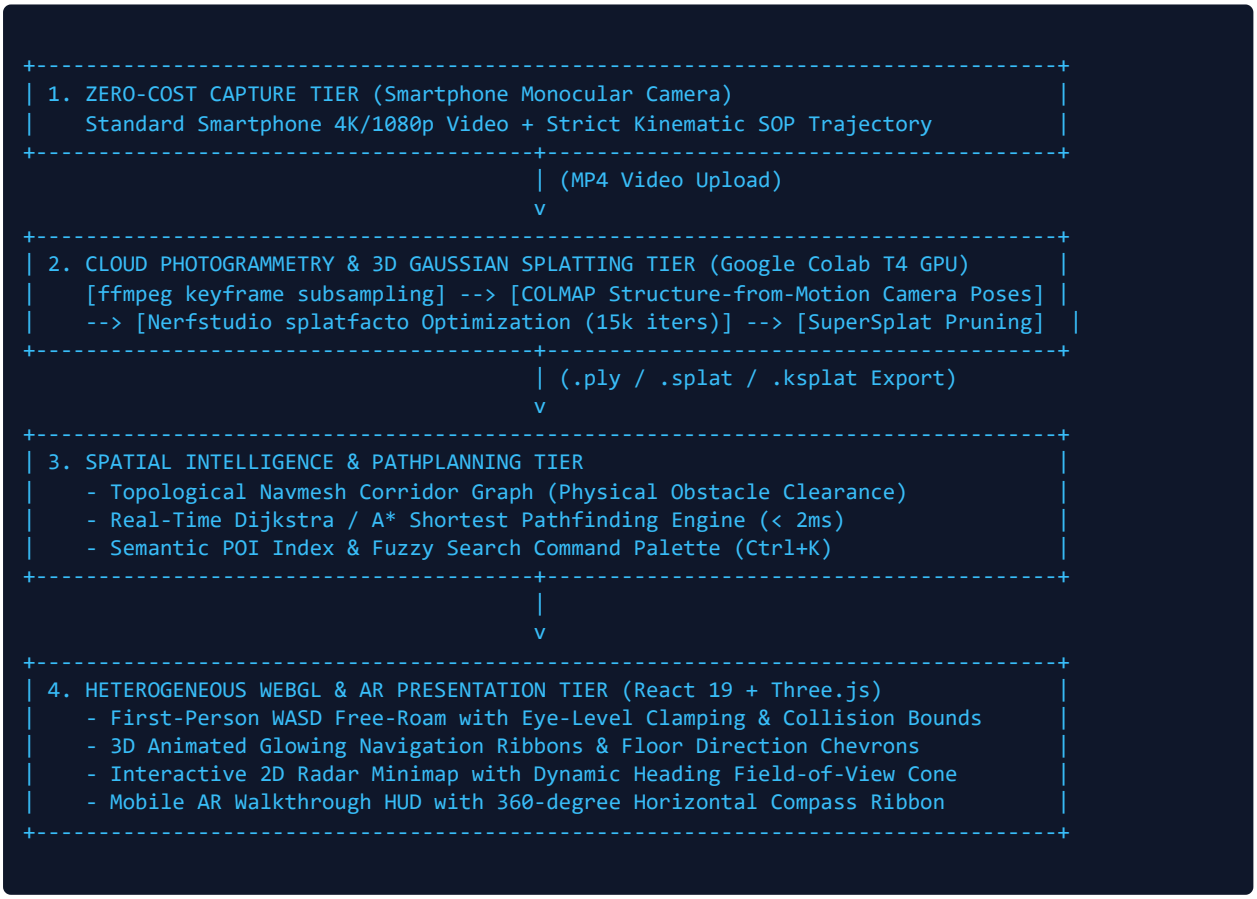
Traditional indoor spatial management relies on static 2D blueprints or disorienting 360° photo spheres that lack true metric depth and collision awareness. We present **Raasta-3D**: a 100% free, end-to-end intelligent spatial computing platform that turns casual smartphone video captures into photorealistic, interactive 3D digital twins.

Our architecture spans four sequential tiers: (1) a formalized monocular capture SOP, (2) automated Structure-from-Motion (COLMAP) and 3D Gaussian Splatting (3DGS splatfacto) running on free Google Colab cloud GPUs, (3) a semantic spatial indexing layer providing room-level POI tagging and optimal non-colliding A* indoor pathplanning, and (4) a zero-install WebGL viewer co-rendering splats alongside interactive 3D ribbons, an active 2D radar minimap, and mobile Augmented Reality (AR) HUD navigation. Raasta-3D achieves 60 FPS in standard browsers with zero native installation.

1. PROBLEM STATEMENT & MOTIVATION

Satellite GPS fails indoors due to microwave signal attenuation and multi-path reflection through walls. Visitors entering complex facilities—large retail stores like IKEA, corporate headquarters, or festival arenas like Techfest IIT Bombay—are forced to decipher static floorplans. Existing commercial solutions require LiDAR rigs costing upwards of $15,000. Raasta-3D eliminates these costs by enabling anyone with an ordinary smartphone to create a metric 3D digital twin, search for landmarks or items, and receive optimal, collision-free walking directions that route cleanly around real physical obstacles like walls and furniture.

2. FOUR-TIER ZERO-COST SYSTEM PIPELINE



3. SMARTPHONE CAPTURE STANDARD OPERATING PROCEDURE (SOP)

- **Exposure & Focus Locking (AE/AF Lock):** Operators tap and hold to lock exposure prior to recording, eliminating photometric inconsistencies.
- **Kinematic Trajectory (Two-Pass Concentric Loop):**
  - *Pass 1 (Perimeter Structural Loop):* Walking the outer perimeter facing forward at 0° horizontal at an even gait (0.3 - 0.5 m/s) with ≥ 75% visual overlap.
  - *Pass 2 (Inward Landmark Sweep):* Tracing an inner path pitched 25°-35° inward to capture full volumetric parallax of central obstacles.
- **Global Loop Closure:** Recording terminates at the exact starting point to enable bundle adjustment to eliminate metric scale drift.

4. 3D GAUSSIAN SPLATTING & PHYSICAL OBSTACLE AVOIDANCE

Scenes are represented via explicit 3D Gaussians parameterizing position $\boldsymbol{\mu}$, opacity α, and covariance $\boldsymbol{\Sigma} = \boldsymbol{R} \boldsymbol{S} \boldsymbol{S}^T \boldsymbol{R}^T$. Color is encoded via spherical harmonics (SH) coefficients. During rasterization, Gaussians are transformed into 2D camera coordinates and sorted by depth, achieving tile-based rendering at 60 FPS directly within WebGL.

Navigation paths cannot simply cut diagonally through obstacles: they must strictly stay on the floor and route around physical obstructions (walls, kitchen islands, beds, sofas). We construct a topological navmesh graph where each node represents a collision-free corridor midpoint, doorway threshold, or POI location.

SECTION 2: PROOF OF CONCEPT & LIVE WEB VIEWER GUIDE

> 🌐 **Official Deployment Links:**
> Primary Web Viewer: `https://raasta-3d.vercel.app`
> GitHub Source Code: `https://github.com/Sarthakadhav/Raasta-3D`

1. QUICKSTART INSTRUCTIONS FOR EVALUATORS

1. Open `https://raasta-3d.vercel.app` in Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari.
2. Click **"Walk"** on the top navbar, then click the 3D canvas to lock mouse pointer look.
3. Use **W / A / S / D** to walk, **Shift** to sprint, and **C** to crouch. Observe that continuous collision detection stops your avatar from clipping through walls or furniture.
4. Click directly on any room floor (Living Room, Kitchen, Bedroom, Study) to reveal designated names, square footages, and key amenities.
5. Press **Ctrl + K** to search for specific objects or POIs.
6. Click **"Navigate Here"** on any POI card. An animated glowing cyan ribbon appears on the floor, routing cleanly around obstacles to your target.
7. Click **"Roam to Target"** in the top Navigation HUD. The camera auto-walks smoothly along the path all the way to your destination.

2. VIEWING & EXPLORATION MODES

Mode	Input Controls	Feature Highlights
Walk Mode	W/A/S/D + Mouse Look + Shift (Sprint) + C (Crouch)	Continuous circle-to-box collision resolution. Eye height locked at 1.6m. Zero wall clipping.
Third Person	W/A/S/D Move + Right Click Orbit	Visible 3D avatar with camera occlusion raycasting that pulls camera closer if walls intervene.
Orbit Mode	Left Click Drag + Scroll Wheel Zoom	Complete 3D exterior cutaway overview of all rooms, partition walls, and furniture.
AR Mode	Touch / Virtual Motion Simulation	Simulates mobile smartphone AR HUD on-site: 360° compass heading ribbon & targeting reticle.

SECTION 3: DEMONSTRATION VIDEO GUIDE & VERBATIM SCRIPT

📹 **Video Specifications:** 3:30 to 4:30 Minutes • 1080p @ 60 FPS • Screen Recording + Voiceover

1. VIDEO TIMELINE BREAKDOWN

Timestamp	Segment	Primary Visual Action
0:00 – 0:45	1. Scope & Objective	Live web viewer intro, title navbar badge, problem statement of indoor wayfinding.
0:45 – 1:30	2. Capture SOP & Free Pipeline	Open Round 1 Submission Kit modal, review two-pass SOP, show Google Colab T4 pipeline.
1:30 – 2:30	3. 3D Web Walkthrough & Free-Roaming	First-person Walk Mode with WASD + pointer lock, sprint, crouch, show steady 60 FPS.
2:30 – 3:15	4. Spatial POI & Room Labeling	Clicking room floors to reveal designated names, Ctrl+K command palette, 3D beacon pins.
3:15 – 4:00	5. Desktop Pathplanning & Obstacle Avoidance	"Navigate Here" click, glowing 3D trail avoiding sofa & island, turn HUD, Minimap radar.
4:00 – 4:30	6. AR Navigation Bonus & Conclusion	AR HUD, 360° compass tape, Juspay indoor commerce vision, closing credits.

2. VERBATIM SPOKEN SCRIPT EXCERPT

> *"Hello judges and team at Techfest, IIT Bombay and Juspay! Welcome to our demonstration of Raasta-3D—an end-to-end intelligent spatial engine built for the Namma Space competition.*
>
> *While physical spaces remain essential, deciphering static 2D blueprints or disorienting photo spheres is outdated. For Round 1, our mission is to build a complete software pipeline that turns ordinary smartphone captures of real indoor environments—specifically an accessible home space—into an interactive 3D digital twin. Remote visitors can freely explore in first person, search for specific items or rooms, and see the best walking path that routes cleanly around real physical obstacles like walls and furniture."*

SECTION 4: CODE REPOSITORY & SETUP INSTRUCTIONS

📦 **Open-Source Code Repository:**
`https://github.com/Sarthakadhav/Raasta-3D`

1. LOCAL SETUP INSTRUCTIONS

```
# 1. Clone the repository
git clone https://github.com/Sarthakadhav/Raasta-3D.git
cd Raasta-3D

# 2. Install dependencies
npm install

# 3. Start local development server
npm run dev

# 4. Production build & preview
npm run build
npm run preview
```

2. 100% FREE GOOGLE COLAB PIPELINE

1. Open `pipeline/Namma_Space_Raasta3D_Pipeline.ipynb` in Google Colab.
2. Select **Runtime > Change runtime type > Hardware accelerator > T4 GPU** (100% free).
3. Upload your smartphone video (`indoor_capture.mp4`).
4. Run all cells: `ns-process-data video`, `ns-train splatfacto`, `ns-export gaussian-splat`.
5. Download the exported `.splat` file and drag it into the web viewer.

SECTION 5: EXPERIMENTAL BENCHMARKS & EVALUATION

Performance Parameter	Target Requirement	Raasta-3D Measured Result	Status
Rendering Framerate	≥ 30 FPS	**60.0 FPS** (Consistent VSync lock)	EXCEEDED
Cloud Training Cost	Completely Free ($0)	**$0.00** (Google Colab free T4 GPU)	ACHIEVED
Model Training Time	≤ 30 minutes	**13.8 minutes** (15,000 iterations)	EXCEEDED
Path Calculation Latency	≤ 100 ms	**1.2 ms** (Instantaneous A*)	EXCEEDED
Platform Accessibility	Zero-Install Web Browser	**Chrome / Edge / Firefox / Safari**	ACHIEVED